

HYEONBEEN LEE

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EDUCATION

Virginia Tech, Blacksburg, VA, USA

Aug. 2026 — May 2031 (Expected)

Ph.D. in Mechanical Engineering (Advisor: [Simon Stepputtis](#))

Research Focus: Robot Learning, Neurosymbolic AI, Physics-aware ML

Kyung Hee University, Seoul, South Korea

Mar. 2022 — Feb. 2024

M.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#))

GPA: 3.93/4.0

Thesis: ‘Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems’

Kyung Hee University, Seoul, South Korea

Mar. 2015 — Feb. 2022

B.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#), [Shinkyu Jeong](#))

(Upper Div.: 3.88/4.0) GPA: 3.52/4.0

Thesis: ‘Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization’

(* Military leave of absence: May 2017 — Feb. 2019)

PUBLICATIONS

Journal Papers

4. **H. Lee**, M. Jung, T.K. Yeu, J.B. Han, D. Park, J.G. Kim (2026), “Frequency-aware decomposition learning for sensorless wrench forecasting on a vibration-rich hydraulic manipulator”, submitted to *IEEE/ASME Transactions on Mechatronics*. [\[arXiv\]](#) [\[code\]](#)
3. B. Koo, M. Jung, **H. Lee**, T. Yeu, J.G. Kim, J.B. Han, Y. Lee, D. Park (2026), “Buoyancy-integrated hybrid reaction force estimation method with real-time haptic feedback for underwater hydraulic manipulation”, submitted to *IEEE Robotics and Automation Letters*.
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), “cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics”, *Journal of Computational Physics (WoS IF Top 2.5% in Physics, Mathematical)*, 496, 112578. [\[link\]](#) [\[code\]](#)
1. S. Han*, G.E. Jeong*, **H. Lee**, W.S. Choi, J.G. Kim (2023), “Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions”, *Nuclear Engineering and Technology (WoS IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. (*Co-first authors) [\[link\]](#)

Conferences & Proceedings

6. J.H. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu. “Digital twin model of underwater construction robot for real-time grinding simulation”, *7th International Conference on Multibody System Dynamics*, Madison, WI, USA, Jun. 2024 [\[link\]](#)
5. **H. Lee**, J. Han, T.K. Yeu, J.G. Kim. “Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer”, *Korean Society of Mechanical Engineers* (Oral Presentation), Incheon, South Korea, Nov. 2023 [\[link\]](#)
4. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation”, *Korean Society of Mechanical Engineers* (**Excellent Paper Award**, Oral Presentation, * Extended works from IMAC 2023), Busan, South Korea, May 2023. [\[link\]](#)
3. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Composite neural network with differentiation propagation for modeling nonlinear impulsive dynamics”, *Conference on Dynamics and Control, Korean Society of Mechanical Engineers* (Oral Presentation), Jeju, South Korea, Mar 2023 [\[link\]](#)
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Composite neural network framework for modeling impulsive nonlinear dynamic responses”, *41st International Modal Analysis Conference (IMAC)* (Oral Presentation), Austin, TX, USA, Feb. 2023 [\[link\]](#)
1. **H. Lee**, J.G. Kim. “Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation”, *Korean Society for Noise and Vibration Engineering* (Oral Presentation), Jeju, South Korea, Dec 2022

AWARDS & HONORS

- **Excellent Paper Award** Korean Society of Mechanical Engineers, 2023
 - Awarded for a first-author presentation at KSME 2023 conference.
- **Brain Korea 21 Plus Scholarship** Graduate School, Kyung Hee University, 2022 — 2023
 - Received \$16K in graduate funding during M.Eng. studies.

- **Graduate Assistantship** Graduate School, Kyung Hee University, 2022 — 2023
 - Received \$11K through a graduate assistantship during M.Eng. studies.
- **Academic Superiority Scholarship** Dept. of Mechanical Engineering, Kyung Hee University, Spring 2021
 - Ranked 1st in the GPA-based departmental merit review and received a full tuition waiver valued at \$3K.

RESEARCH EXPERIENCES

Graduate School, Kyung Hee University Research Assistant
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Mar. 2024 — Apr. 2026

- Developed a frequency-aware decomposition learning method for robotic wrench forecasting. Submitted a first-authored manuscript to the *IEEE/ASME Transactions on Mechatronics (T-MECH)*.
- Refined the forecasting model for real-time robotic wrench estimation for [KRISO](#).

Kim Jaechul Graduate School of AI, KAIST Research Intern
Cognitive Learning for Vision & Robotics Lab., advised by *Joseph J. Lim* Sep. 2023 — Nov. 2023

- Conducted an in-depth review of [skill-prior regularized reinforcement learning \(SPiRL\)](#) and delivered a 90-minute presentation to lab members and faculty.

Graduate School, Kyung Hee University Graduate Research Assistant
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Mar. 2022 — Feb. 2024

- Proposed a composite neural network architecture for learning complex dynamics from multi-order derivatives. Published a first-author article in the *Journal of Computational Physics (JCOMP)*.
- Developed a multibody dynamics vehicle model for the [Korea Atomic Energy Research Institute \(KAERI\)](#). Published a co-authored article in the *Nuclear Engineering and Technology (NET)*.
- Designed a learning-based model to forecast the contact-rich wrench of a hydraulic manipulator for the [Korea Research Institute of Ships and Ocean Engineering \(KRISO\)](#).

College of Engineering, Kyung Hee University Undergraduate Research Assistant
Modeling & Simulation Lab., advised by *Jin-Gyun Kim* Jan. 2021 — Feb. 2022
Aerodynamic Design Exploration & Big Data Analysis Lab, advised by *Shinkyu Jeong* Sep. 2020 — Jun. 2021

- Investigated data-driven methods for flexible multibody dynamics and aerodynamics.

SKILLS

- **Programming Languages:** Python · MATLAB · C++ · L^AT_EX
- **Software & Frameworks:** PyTorch · Git · Docker · ROS · RecurDyn · Gazebo · MuJoCo · OpenAI API
- **Scientific Computing:** Flexible multibody dynamics · Finite element methods · Numerical analysis
- **Natural Languages:** Korean (Native) · English (TOEFL iBT 111/120) · Japanese (Advanced proficiency)

INVITED SEMINARS

2. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, IMBD Lab, Chungnam National University, Daejeon, South Korea, Apr. 16, 2026 [\[keynote\]](#)
1. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, Modeling & Simulation Lab, Kyung Hee University, Yongin, South Korea, Apr. 14, 2026 [\[keynote\]](#)

SELECTED PROJECTS

- Segment-Anybody, at *Dept. of AI, Kyung Hee University* [\[code\]](#)
 - Fine-tuned Meta’s Segment-Anything Model (SAM) for detailed human body parts segmentation in human-robot interaction.
- Deep seq2seq from scratch, *Personal project*. [\[code\]](#)
 - Scratch-implemented deep sequence-to-sequence models, from RNN to Transformer variants.
- ROS SLAM on Apple Silicon devices, *Personal project*. [\[code\]](#)
 - Addressed ROS TurtleBot3 SLAM and Gazebo compatibility issues on the Apple Silicon MacOS devices.
- RecurDyn ProcessNet with Python, at *Modeling & Simulation Lab*. [\[code\]](#)
 - Automated and parallelized flexible multibody dynamics (FMBD) simulations of RecurDyn.

TEACHING & MENTORSHIP

Research Mentor Modeling & Simulation Lab., Kyung Hee University
— Guiding junior graduate students on robotics and machine learning. Mar. 2025 — Present

Teaching Assistant, ME380 Systems Dynamics Dept. of Mechanical Engineering, Kyung Hee University
— Lectured on multibody dynamics with theoretical derivation and its computational implementation. Mar. 2022 — Jun. 2023

COMMUNITY SERVICE

Lead Administrative Assistant	Dept. of Mechanical Engineering, Kyung Hee University, 2022 — 2024
48th College of Engineering Student Council	Kyung Hee University, Feb 2019 — Jan. 2020
ROK-U.S. Combined Marine Corps Interpreter	1st Marine Div., ROKMC, Sep 2017 — Feb. 2019

REFERENCES

- **Jin-Gyun Kim**, Ph.D.
Associate Professor at Department of Mechanical Engineering, Kyung Hee University (jingyun.kim@khu.ac.kr)
- **Hee-Sun Choi**, Ph.D.
Assistant Professor at Department of Artificial Intelligence and Robotics, Sejong University (heesunchoi@sejong.ac.kr)
- **Seongji Han**, Ph.D.
Assistant Professor at Department of Mechatronics Engineering, Chungnam National University (seongji.han@cnu.ac.kr)